

Reading Prosody: A Listening Guide for Teachers

Leilani Sáez, Makayla Whitney, Joseph F. T. Nese, Julie Alonzo, Rhonda N. T. Nese

Teachers can enhance students' oral reading fluency by fine-tuning how they listen to three reading prosody components: intonation, amplitude, and timing.

Years ago, Hudson et al. (2005) wrote a seminal article that described how to assess reading fluency using multiple measurement indicators. Since then, accuracy and speed indicators have become common reading fluency measures used in classrooms, but prosody indicators have not (Ardoín et al., 2013; Benjamin et al., 2013; Rasinski, 2012). The field has been slow to incorporate prosody into reading fluency practice because its complexities are still poorly understood, despite its intuitive appeal.

Long described as “reading expression” (Kuhn & Stahl, 2000), prosody’s specific role in reading fluency is not well known. For example, as part of a research project, we trained 56 educators across 20 U.S. states to rate children’s reading prosody using a four-level systematic rubric (adapted from the National Assessment of Educational Practice [NAEP] and Multidimensional Fluency Scale [MDFS] tools; see the Take Action! Sidebar). The educators had, on average, 10 years of experience. Yet, they described widely ranging levels of prior knowledge about prosody’s role in reading. Before working on our project, only 24% of these classroom teachers, reading specialists, and administrators confidently knew about the term *reading prosody*.

The educators in our project are not alone. In scientific studies, prosody ratings for the same oral reading passages have been low to moderately correlated (Haskins & Aleccia, 2014; Morrison & Wilcox, 2020), suggesting a persistent uncertainty about how to evaluate reading prosody, even among extensively trained raters. As part of our project, educators completed hours of training that resulted in multi-step certification (evaluating, on average, 160 oral reading fluency audio recordings of students across second–fourth grades; Nese et al., 2020). All 56 educators completed a survey and 52% reported experiencing meaningful challenges with evaluating students’ reading prosody. They especially found fine-grained nuances that they heard, such as shifts in a reader’s pace or pitch, hard to account for when determining overall scores.

Yet, these project educators also reported learning to “fine-tune” their listening over time. They described expanding their understanding of “how reading can sound.” Although experienced in listening to students read, they had previously lacked specific knowledge about the meaning of different prosody indicators to guide their focus and evaluation. Once educators knew *what* to listen for and *why* it mattered, they could analyze students’ reading fluency more strategically.

Our goal is to share the research about reading prosody and clarify its specific role in developing students’ reading fluency beyond the general recognition that expression matters. Throughout this paper, we note three pseudonymed readers from our project: Fast Faith, Choppy Chase, and Struggling Stella. At the time that their oral readings were recorded, Faith and Stella were in second grade and Chase was in third grade; all three readers were monolingual English-speaking students attending elementary schools in the Pacific Northwest. Their low scores on our reading prosody rubric suggested different instructional support needs to improve their reading fluency. We share their cases to illustrate how teachers can more comprehensively strengthen students’ fluency, beyond accuracy

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and rate, by intentionally listening for and addressing specific issues with reading prosody.

What is Prosody?

Often characterized as “the music of language” (Hudson et al., 2005, p.704), prosody includes the patterns of rhythm, stress, and tone of speech. When used to analyze reading quality, prosody provides valuable information beyond how successfully a sentence’s literal meaning has been decoded. For example, fluent reading demonstrates a correct recognition of printed text spoken with a pace and melodic flow resembling natural conversation (Hudson et al., 2005; Schwanenflugel et al., 2004). So, along with accurate word recognition and efficient reading rate, prosody is a fundamental driver of fluency that supports reading comprehension (see Figure 1). Accuracy describes the correctness of the reading and rate describes its speed. Prosody reveals how well readers integrate and glean meaning while reading. For teachers, it provides

an important bridge for understanding the quality and strength of connections that readers make. A fluent reader not only correctly recognizes words efficiently, but also generates relevant meaning in the process.

Reading comprehension relies on both word-reading ability and prosody (Kim et al., 2021; Miller & Schwanenflugel, 2008). As reading develops in the first–

ninth grades, prosody plays a moderately significant role ($r = .51$; Wolters et al., 2022). However, its importance shifts over time. Before third grade, reading prosody inconsistently predicts children’s comprehension (Schwanenflugel et al., 2004). But by fifth grade, it reliably predicts decoding and comprehension skills, eventually becoming more strongly linked with comprehension than word recognition (Veenendaal et al., 2016). This growing association holds even for silent reading (Paige et al., 2014). In some studies, prosody expressiveness has explained

comprehension skill levels beyond rate and accuracy indicators (e.g., Benjamin et al., 2013; Tong et al., 2022), highlighting its unique contribution to fluent reading. Exactly how these three fluency drivers support each other across reading development and scaffold comprehension remains unspecified (as discussed by Ashby, 2016; see also Schwanenflugel et al., 2004), so there is still much to learn.

Nevertheless, researchers generally agree that fluent oral reading should sound like speech (Dowhower, 1991; Godde et al., 2020; Stahl & Kuhn, 2002). As the “basic cognitive skeleton that allows one to hold an auditory sequence in working memory” (Kuhn et al., 2010, p.235), prosody involves both receptive (internally processed) and expressive (vocalized) aspects of oral language that influence reading skill (Lochrin et al., 2015). So, the strength of students’ expressive oral reading depends on the strength of their receptive listening capabilities. For example, young children’s sensitivity to aspects of spoken prosody predicts their phonological awareness (Goswami et al., 2010), which is crucial to the prompt development of later decoding skills (Ehri, 2020).

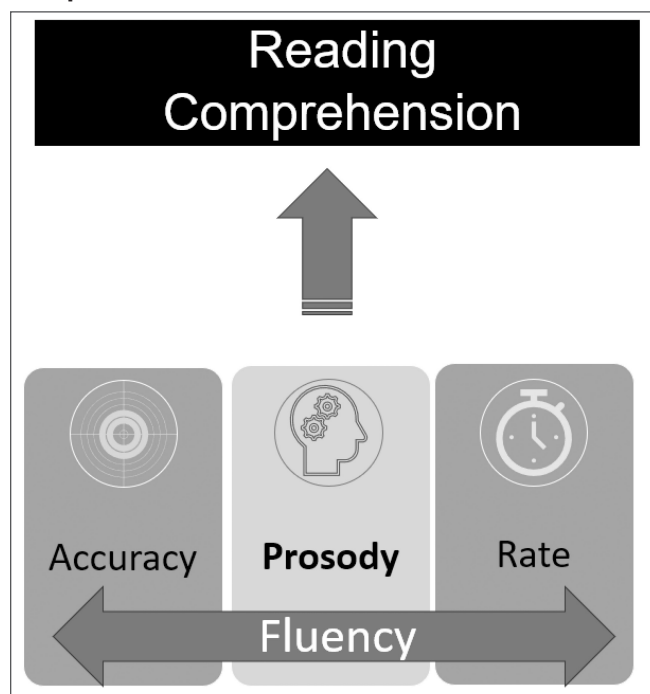
Prosody’s Three Main Components

Prosody can be examined at multiple text levels: word (e.g., through phonetic or syllabic stress patterns), phrase (e.g., through syntax organization), and discourse (e.g., through the accentuation of certain sentence parts that are

PAUSE AND PONDER

- How might you use aspects of prosody intonation, amplitude, or timing to assess oral reading fluency differently?
- What is something you learned about reading prosody that can be incorporated into your classroom reading instruction?
- How might you talk with parents or other educators differently about the development of reading fluency?

Figure 1
Fluency Elements and their Relation to Reading Comprehension



surprising, new, or dramatic; Wade-Woolley et al., 2022). Over the past two decades, three key prosody components have been studied across these levels: *intonation*, *amplitude*, and *timing* (e.g., Holliman et al., 2014; Kim et al., 2021; Schwanenflugel et al., 2004). We describe each below.

Intonation

Intonation refers to how readers vary their pitch while reading, as measured by sound wave frequency. It is often characterized as the different voice inflections heard during oral readings. It includes both a recognition of the words (i.e., lexical prosody; Schwanenflugel & Benjamin, 2017) and their interpretation (Wade-Woolley et al., 2022). Some researchers further distinguish between intonation and expressivity (e.g., Erkeson, 2010; Godde et al., 2020). To them, intonation reflects voice changes made in response to word and grammatical aspects of a text while expressivity reflects a person's mentally constructed interpretation of meaning while reading.

In general, readers shift their intonation based on their own internalized rules for reading—the extent to which they have mapped speech tendencies to printed text. However, certain features of text can also influence prosody. For example, by second grade, typical English-language readers become increasingly competent at letting their voices fall at the end of sentences; they understand that a period represents a breakpoint in the stream of words they are reading (Miller & Schwanenflugel, 2006; Schwanenflugel et al., 2004). In contrast, readers less consistently change their pitch with commas (Miller & Schwanenflugel, 2006), suggesting less certainty about their purpose. And contrary to common belief, typical readers use *both* falling and rising pitches when reading questions (Miller & Schwanenflugel, 2006); neither characterizes better reading.

Intonation differences can also naturally show up in the way that specific words are read (Kuhn et al., 2010). How syllables and other word parts are stressed when reading depends on how spoken words are stressed (Goetry et al., 2006). For example, students might differ when reading suffixes (e.g., reading “COLDest” versus “coldEST”), compound words (e.g., reading “RAINbow” versus “rainBOW”), or various parts of speech (e.g., reading “use” as a noun with the /s/ sound versus as a verb with the /z/ sound). Even where students live can change how words are read (Schwanenflugel et al., 2004). However, regional or dialectal speech differences are only a reading prosody concern when they disrupt students' process of meaning-making.

In addition, Erikson (2010) argues that expressivity in readers' prosody reveals something else for teachers to notice: what students are inferring at an even deeper,

discourse level of understanding. How good readers use their voices to stress italicized words functions similarly to how good speakers accentuate spoken words for emphasis. So, expressive prosody reveals what readers are attending to and how they are thinking about the text (e.g., What are they envisioning in their minds as they read? What personal connections are being made?). However, teachers need to balance expectations of reading expressiveness with students' speaking expressiveness. Children who do not speak with a wide range of pitch or emphasis won't likely read with one unless specifically instructed to do so (Godde et al., 2020). The educators in our project shared how prosody offered them more clues to students' “internal dialogue” that often reflected their lived experiences. For example, how certain parts of passages were read uniquely revealed students' level of text engagement. As one educator noted, “People who read with expression know what is happening in the story and what voice to give the reading.” In other words, readers are doing more than deciphering words; expressive reading is more than melodic to hear, it is meaning-filled. Therefore, beyond listening for reading that sounds like natural speech, readers' intonation can shed light on their understanding of what is being read.

Amplitude

Amplitude refers to sound wave volume or intensity. Although changes in overall loudness can reveal a reader's comfort or effort level (e.g., whispered mumbles or poor volume regulation may reflect working memory limitations on sentence processing; Schremm et al., 2015), contemporary research in this area is virtually non-existent (although see Breznitz & Leikin, 2001). However, mind-wandering while reading aloud is associated with greater voice amplitude (and minimized pitch variability; Franklin et al., 2014). Students who read automatically, but sound flat (i.e., they stop controlling their voices) are cognitively “zoning out”; basic word recognition is occurring but without much comprehension. So, this is a sign to re-direct readers back to the task of reading to generate meaning. On the other hand, flat reading can also suggest that decoding has “maxed out” a reader's ability to engage in other aspects of cognitive processing (e.g., self-monitoring or inferencing). One of the educators in our project noted how learning about these reading prosody differences changed her perspective on monotone reading; she now recognizes it as a sign that comprehension may be suffering and something for her to immediately address.

Subtle amplitude differences that distinguish between readers (e.g., slight verbal accentuations) may not be easily perceived by listeners but can be captured through

technology. Computerized amplitude analysis, for example, can efficiently reveal such fine-grained distinctions. Approaches using headsets, recordings, and sophisticated software are becoming increasingly accessible in classrooms and potentially offer teachers more useful information.

For example, technology tools may aid early intervention efforts by indicating speech processing issues that impact reading prosody. Neural difficulties in discriminating between fine-grained speech information, like phonemes, would be easier to swiftly detect (Hämäläinen et al., 2013; Ríos-López et al., 2017). This matters because when tiny, but significant, bits of auditory information go undetected, readers become unable to “lock” their attention into sound wave cues that help them anticipate where speech is headed (and how/when text will end; Cutler & Darwin, 1981; Grosjean, 1983). This makes memory retrieval harder and disrupts the mental coordination needed to process phonetic, lexical, and syntactic aspects of language for reading (Myers et al., 2019). Therefore, amplitude analysis of reading may someday illuminate early speech and processing problems to help prevent persistent reading difficulties like Dyslexia, in which speech amplitude processing is weak (Goswami et al., 2016).

Timing

Timing refers to the rhythm and pace of reading, as measured in syllables per second. It is often considered the phrasing, smoothness, and tempo of reading—how a reader stops and starts (Schwanenflugel et al., 2004). Both *duration* (how long reading advances without hesitation, and the length of pauses) and *frequency* (how often words are repeated or re-read, and how often long pauses interrupt reading flow) reveal the quality of a reader's timing. Ideally, one's oral reading rate should match their conversational speaking rate (Young & Rasinski, 2018). Many of the educators in our project expressed concern about the “overstressing” of fast reading at their schools. Although often credited as evidence of fluent reading, a teacher's ability to determine what a student comprehends is undermined when reading is done for speed instead of for understanding.

Timing is often analyzed in segments, such as the boundaries between pauses (Cowie et al., 2002), to clarify how reading is flowing. Sophisticated readers use well-placed pauses that coincide with sentence grammar (Benjamin & Schwanenflugel, 2010). Good readers make fewer and shorter breaks in their reading (Schwanenflugel et al., 2004), resulting in a smooth and steady flow. Decoding plays a crucial role in timing; readers with better decoding skills produce shorter pauses between

sentences compared to those with weaker decoding skills (Schwanenflugel et al., 2004). However, the length of time it takes to sound out words is not the only crucial factor.

A reader's timing can also be affected by the size and number of “chunks” used to organize sentences into meaningful syntactic phrases (Schreiber, 1991), which is partly influenced by working memory capacity (Bishop, 2021; Kuhn et al., 2010). Students with greater working memory capacity can cognitively process more text, enabling them to read longer phrases within sentences in a passage more smoothly. Reading flow is more disrupted for students with limited working memory capacity because they must process smaller amounts of text that impair their sentence-level meaning-making. This kind of timing pause is different from a hesitation resulting from decoding problems. When readers lack cohesion—within and between sentences—their ability to understand the big picture often leads to confusion (and also hampers their ability to anticipate what will happen next in the text).

What Can Prosody Tell us about Supporting Struggling Readers?

Based on their review of the research, Godde et al. (2020) concluded that reading prosody primarily develops between the first and fifth grades, with pitch timing refinements continuing into adulthood. By late elementary, typical reading prosody becomes more adult-like (Álvarez-Cañizo et al., 2015; Schwanenflugel et al., 2004). The prosody of struggling readers, however, reveals persistent and specific patterns of difficulty across prosody components that teachers can listen for and address across most grades. Below and in Table 1, we discuss some student issues and instructional recommendations.

Common reading prosody issues with intonation involve how students use their voices (e.g., the appropriateness of word-sound emphasis, pitch changes, and other forms of expression). In our project, Fast Faith used the same pitch to read an entire passage. She lacked expression and observable meaning-making as she read as fast as she could. In contrast, Choppy Chase and Struggling Stella varied their tones in response to uncertainties they experienced while reading. Both raised their pitch when reading words they didn't immediately recognize, making a “scooping” (upward rising) sound. Stella's voice often sounded like a wave of frequent within-word rising and falling, reflecting her significant challenges with word decoding. At one point, Choppy Chase's misunderstanding of the text was clear from an unexpected change in expression. He abruptly misread the middle of a declarative sentence as an emphatic command (instead of simply finishing the sentence with a more neutral

Table 1
An Analysis of Three Reading Prosody Cases and their Suggested Instructional Supports

Reader	Reading prosody components			Suggested instructional supports
	Intonation	Amplitude	Timing	
Fast Faith (Rating 3)	No pitch changes throughout	Same loud voice heard throughout	Rapid reading without pauses throughout	■ Slow down reading to her typical speaking pace to support more meaning-making ■ Lessen the reading demands to enable self-monitoring of her volume and engagement
Choppy Chase (Rating 2)	Uneven pitch for uncertain words (e.g., <i>twin</i> had upward scooping)	Natural volume except for a very loud final sentence	Reading disjointed throughout: frequent repetitions, self-corrections (e.g., <i>bracelet</i> read as “brak-brake-let”), lengthy hesitations	■ Reinforce reading larger portions of connected text to generate big-picture meaning (beyond word-level decoding) ■ Use think-alouds and peer reading to provide natural speech-like models
Struggling Stella (Rating 1)	Frequent within-word pitch changes (e.g., <i>day</i> had high-low-high rises & falls)	Whispered and inaudible words heard throughout	Sentences sounded like disconnected words: long hesitations, unfamiliar words sounded out slowly (“ffrrrom... from”); unknown words skipped	■ Use easier texts to strengthen word recognition fluency ■ Practice self-monitoring to understand what is read

tone). The connection he made, although meaningful, was based on an incorrect reading of the text.

Although some amplitude issues can be harder to detect in the classroom, an easily observable concern involves poor volume regulation. For example, Fast Faith read the entire passage loudly and intensely; her focus on accuracy and speed left her unable to monitor and control her voice volume while she read. In contrast, Struggling Stella strongly controlled her volume by whispering words she didn’t know throughout the text, making it hard to evaluate the accuracy of her word recognition. Choppy Chase mumbled through unfamiliar words that resulted in occasionally softened volume that mirrored his word recognition uncertainty, not expressiveness. Despite their differences, these reading behaviors all call for a teacher’s attention; they reflect symptoms of reading goal, decoding, and meaning-making problems that sabotage the development of fluent reading.

Timing issues can also show up differently across struggling students. Although readers might begin oral passage readings with similar timing, they can diverge as the reading progresses (Cowie et al., 2002). Weaker readers tend to become more piece-meal in their phrasing, especially as page-turning or challenging punctuation disrupts their processing (e.g., parentheses and commas can slow them

down; Cowie et al., 2002). They are also likely to repeat or inaccurately read complex words, as well as pause longer, which creates a sense of uneven timing that sounds “clunky” or awkward (Alves et al., 2015; Binder et al., 2013). For example, Choppy Chase’s reading was disjointed, both within and between sentences as he strived to decode. Although short phrases were evenly paced, he took unusually large breaths (creating long silent pauses) before beginning new sentences as his brain coordinated what he just read with what he would read next. Lengthy phrases contained unmeaningful breaks that sometimes led to misunderstood chunks of text. In contrast, Struggling Stella delayed the start of most sentences and frequently skipped words; she also rushed through words she recognized and repeated phonemes to “sound them out”, resulting in uneven pacing. Her frequent hesitations made the reading sound more like isolated words than connected text. Fast Faith’s reading, on the other hand, lacked pauses of any kind, sounding like a single, long stream of words (that offered no evidence of understanding).

Educator Takeaways

The educators in our project strongly felt that a better understanding of reading prosody can enhance teachers’

fluency assessment and instruction practices. One of the educators shared, “As a teacher, it’s extremely beneficial to know that students can read with good prosody, even if their actual reading fluency word count [on an oral reading fluency assessment] is low.” Some reported that knowing more about reading prosody helped them to think about the development of reading skills more holistically by shifting their thinking about what fluency should sound and where growth opportunities could be found across diverse levels of reading ability. As one educator remarked, “If students know what they should sound like they can work towards it.”

Instructional Support

Educators reported learning how small, but potent, distinctions in intonation helped to clarify signs of difficulty that had been previously overlooked. Multiple educators characterized prosody as “an example of engagement with text.” Flat intonation and dramatic changes in pitch became features to not only notice but also get curious about, to better grasp students’ text engagement. Whereas hearing students read with a melodic flow may have been teachers’ general goal in the past, they came away from the project with a more nuanced understanding of good versus concerning aspects of intonation to listen for and evaluate.

In addition, educators learned how reading pauses represented something more than just silence; analyzing the underlying reasons for observed pauses became a new goal, as well. For example, weak decoding or difficulties in anticipating sentence structure might explain frequent disruptions to reading flow, like in the case of *Choppy Chase*.

But long and widespread timing gaps, like in the case of *Struggling Stella*, might instead suggest the need to use an easier text to properly evaluate prosody (texts that are too challenging obscure how well students can generate meaning from print they recognize). Even a lack of pauses, like in the case of *Fast Faith*, invites teacher reflection about refining students’ reading prosody and understanding to improve their fluency.

Diagnostic Support

The surveyed educators also shared how applying reading prosody more directly in fluency assessment would provide “valuable insight into how well a student is progressing in reading”. For example, educators noted that reading prosody enables them to gather more well-rounded information beyond Words Correct Per Minute (WCPM), which they believed creates a better understanding of student comprehension gaps and specific needs to target for bolstering fluency. They also expressed how incorporating prosody components into their fluency assessments would enable them to “go beyond” numerical scores in discussions with parents, other teachers, and students. Many envisioned that their learning about reading prosody would facilitate richer descriptions of students’ overall reading proficiency moving forward.

Most educators in our project felt that using a systematic rubric or numeric scale aided their ability to capture and compare specific reading prosody features across students and over time. Being able to recognize, with greater precision than a quick impression, gave them a stable marker to inquire about and strengthen. A few educators even noted how audio recordings and scores could

Table 2
Prosody Features to Listen for When Supporting Struggling Readers

Considerations	Intonation	Amplitude	Timing
Also Known As	Expression Pitch	Intensity Volume	Phrasing Rhythm
Useful for Understanding	Word recognition skill Readers’ interpretation	Engagement/effort	Text boundaries understanding Decoding skill
Signs of a Struggling Reader	Inaccurate vocal stress Minimal or context-inappropriate changes	Overly loud or quiet voice	Awkward flow Long & frequent pausing/hesitations
Listener Questions to Ask Yourself	Does emphasis or expression reflect proper word recognition or meaning-making? Are there language or cultural differences to consider?	Why does the sound seem stuck at one level? Is there cognitive overload (from text difficulty or surroundings?)	How appropriate are pauses or hesitations—what is causing them? When is good phrasing heard?

be used in digital portfolios to track progress (within and across grades) in the future.

Although limited, a few prosody rubrics that include *intonation* and *timing* elements are available for classroom use (see the Take Action! sidebar and More to Explore links). Two well-known fluency scoring rubrics are the NAEP fluency scale (Daane et al., 2005) and the Multidimensional Fluency Scale (MDFS; Rasinski et al., 2009). A more recent version of the NAEP reading fluency assessment includes the use of an eight-level expression rubric for scoring proficiency (White et al., 2021). This newer tool assesses expression, ranging from “word by word” to full “passage expression”, reflecting how reading prosody progresses along a continuum much like the other two indicators of fluency. We believe that tools like these offer teachers a more comprehensive way to promote students’ reading fluency.

Conclusion

Through their learning about reading prosody, educators in our project more easily noticed and appraised aspects of students’ intonation, amplitude, and timing to “shed light beyond what students can’t do.” They reported becoming more aware of how the quality of oral reading connects to text engagement and comprehension. As one noted, “[Now] I can gauge where students are along the prosody scale and know how to teach it as well.” Across their survey responses, educators appreciated how their newfound understanding clarified and widened the focus on reading fluency development.

The best practices for supporting reading prosody instruction are still under study, however. For example, common fluency-boosting tactics, such as story re-reading and modeling good prosody (Young et al., 1996) may be inadequate; more intentional strategies may be needed. However, as a starting point, Young et al. (2015) found that reading expression, rate, and accuracy were enhanced when students experienced modeling, engaged in meaningful practice, and received feedback from a tutor for approximately 20 min for 1 month. The key program features to model and practice, as well as the best feedback tactics to use class-wide, have yet to be firmly established—more work still needs to be done.

One promising assessment approach that can support instruction comes from the work of Rebekah Benjamin et al. (2013), who created a simple rubric for assessing fluency more comprehensively. Their four-level rating scale combines WCPM with *intonation* (pitch changes, flatness) and pausing (e.g., repetition and hesitation timing) elements of prosody, both of which are relevant and feasible to address through classroom instruction. We

TAKE ACTION!

Try using our project-developed reading prosody scoring rubric to extend oral reading fluency scores. Consider the differences between emergent (1) to proficient (4) proficiency levels. Use the descriptions to help you listen for the quality of their passage-reading intonation/expressiveness, amplitude/phrasing and timing/pace (adapted from the well-known NAEP and Multidimensional Fluency Scales).

Score	Description
4	Reads primarily in larger, meaningful phrase groups. Although some regressions, repetitions, and deviations from text may be present, these do not detract from the overall structure of the story. Reads smoothly with some breaks but self-corrects with difficult words and/or sentence structures. Some or most of the story is read with expressive interpretation. Reads with varied volume and expression (like talking to a friend with voice matching the interpretation of the passage).
3	The majority of phrasing seems appropriate and preserves the syntax of the author. Reads with a mixture of run-ons, mid-sentence pauses for breath, and some chopiness. There is reasonable stress and intonation. Little or no expressive interpretation is present.
2	Some word-by-word reading may be present. Word groupings may seem awkward and unrelated to larger context of sentence or passage. Reads moderately slowly. Overall meaning of the text is preserved. Little or not expressive interpretation is present.
1	Read primarily word-by-word. Occasional two-word or three-word phrases may occur—but these are infrequent and/or they do not preserve meaningful syntax. Reads slowly and laboriously. Story line is incoherent. No expressive interpretation is present.

From Nese et al. (2020); adapted from Daane et al. (2005) and Rasinski et al. (2009).

also share our project’s prosody rubric (see Take Action!) and some guiding questions and signs in Table 2 to help teachers strategically analyze their students’ reading prosody straightaway. As one educator noted, learning about reading prosody, “... gives a much more accurate read on the whole aspect of the student as a reader.” Having a greater understanding of intonation, amplitude, and timing prosody aspects of students’ reading offers a new level of meaning to what is heard.

Conflicts of Interest

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ENDNOTE

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REFERENCES

- Álvarez-Cañizo, M., Suárez-Coalla, P., & Cueto, F. (2015). The role of reading fluency in children's text comprehension. *Frontiers in Psychology*, 6, 1–8.
- Alves, L. M., Reis, C., & Pinheiro, A. (2015). Prosody and reading in dyslexic children. *Dyslexia*, 21(1), 35–49.
- Ardoin, S. P., Morena, L. S., Binder, K. S., & Foster, T. E. (2013). Examining the impact of feedback and repeated readings on oral reading fluency: Let's not forget prosody. *School Psychology Quarterly*, 28(4), 391–404.
- Ashby, J. (2016). Why does prosody accompany fluency? Reconceptualizing the role of phonology in reading. In A. Khateb & I. B. Kochova (Eds.), *Current insights from neuro-cognitive research and intervention studies*. Springer International Publishing.
- Benjamin, R. G., & Schwanenflugel, P. J. (2010). Text complexity and oral reading prosody in young readers. *Reading Research Quarterly*, 45(4), 388–404.
- Benjamin, R. G., Schwanenflugel, P. J., Meisinger, E. B., Groff, C., Kuhn, M. R., & Steiner, L. (2013). A spectrographically grounded scale for evaluating reading expressiveness. *Reading Research Quarterly*, 48(2), 105–133.
- Binder, K. S., Tighe, E., Jiang, Y., Kaftanski, K., Qi, C., & Ardoin, S. P. (2013). Reading expressively and understanding thoroughly: An examination of prosody in adults with low literacy skills. *Reading and Writing*, 26, 665–680.
- Bishop, J. (2021). Exploring the similarity between implicit and explicit prosody: Prosodic phrasing and individual differences. *Language and Speech*, 64(4), 873–899.
- Breznitz, Z., & Leikin, M. (2001). Effects of accelerated reading rate on processing words' syntactic functions by normal and dyslexic readers: Event related potentials evidence. *The Journal of Genetic Psychology*, 162(3), 276–296.
- Cowie, R., Douglas-Cowie, E., & Wichmann, A. (2002). Prosodic characteristics of skilled reading: Fluency and expressiveness in 8–10-year-old readers. *Language and Speech*, 45(1), 47–82.
- Cutler, A., & Darwin, C. J. (1981). Phoneme-monitoring reaction time and preceding prosody: Effects of stop closure duration and of fundamental frequency. *Perception & Psychophysics*, 29, 217–224.
- Daane, M. C., Campbell, J. R., Grigg, W. S., Goodman, M. J., & Oranje, A. (2005). Fourth-grade students reading aloud: NAEP 2002 special study of oral reading. The nation's report card (NCES 2006-469). Washington, DC: U.S. Department of Education, Institute of Education Sciences.
- Dowhower, S. L. (1991). Speaking of prosody: Fluency's unattended bedfellow. *Theory Into Practice*, 30(3), 165–175.
- Ehri, L. C. (2020). The science of learning to read words: A case for systematic phonics instruction. *Reading Research Quarterly*, 55, S45–S60.
- Erikson, J. A. (2010). Prosody and interpretation. *Reading Horizons: A Journal of Literacy and Language Arts*, 50(2), 80–98. Retrieved from https://scholarworks.wmich.edu/reading_horizons/vol50/iss2/3
- Franklin, M. S., Mooneyham, B. W., Baird, B., & Schooler, J. W. (2014). Thinking one thing, saying another: The behavioral correlates of mind-wandering while reading aloud. *Psychonomic Bulletin & Review*, 21, 205–210.
- Godde, E., Bosse, M. L., & Bailly, G. (2020). A review of reading prosody acquisition and development. *Reading and Writing*, 33(2), 399–426.
- Goetry, V., Wade-Woolley, L., Kolinsky, R., & Mousty, P. (2006). The role of stress processing abilities in the development of bilingual reading. *Journal of Research in Reading*, 29(3), 349–362.
- Goswami, U., Cumming, R., Chait, M., Huss, M., Mead, N., Wilson, A. M., Barnes, L., & Fosker, T. (2016). Perception of filtered speech by children with developmental dyslexia and children with specific language impairments. *Frontiers in Psychology*, 7, 791.
- Goswami, U., Gerson, D., & Astruc, L. (2010). Amplitude envelope perception, phonology and prosodic sensitivity in children with developmental dyslexia. *Reading and Writing: An Interdisciplinary Journal*, 23(8), 995–1019.
- Grosjean, F. (1983). How long is the sentence? Prediction and prosody in the on-line processing of language. *Linguistics*, 21, 501–529.
- Hämäläinen, J. A., Salminen, H. K., & Leppänen, P. H. T. (2013). Basic auditory processing deficits in dyslexia: Systematic review of the behavioral and event-related potential/field evidence. *Journal of Learning Disabilities*, 46(5), 413–427.
- Haskins, T., & Aleccia, V. (2014). Toward a reliable measure of prosody: An investigation of rater consistency. *International Journal of Education and Social Science*, 1(5), 102–112.
- Holliman, A. J., Williams, G. J., Mundy, I. R., Wood, C., Hart, L., & Waldron, S. (2014). Beginning to disentangle the prosody-literacy relationship: A multi-component measure of prosodic sensitivity. *Reading and Writing*, 27, 255–266.
- Hudson, R. F., Lane, H. B., & Pullen, P. C. (2005). Reading fluency assessment and instruction: What, why, and how? *The Reading Teacher*, 58(8), 702–714.
- Kim, Y. S. G., Quinn, J. M., & Petscher, Y. (2021). Reading prosody unpacked: A longitudinal investigation of its dimensionality and relation with word reading and listening comprehension for children in primary grades. *Journal of Educational Psychology*, 113(3), 423–445.
- Kuhn, M. R., Schwanenflugel, P. J., & Meisinger, E. B. (2010). Aligning theory and assessment of reading fluency: Automaticity, prosody, and definitions of fluency. *Reading Research Quarterly*, 45(2), 230–251.
- Kuhn, M. R., & Stahl, S. A. (2000). Fluency: A review of developmental and remedial practices. (Report No. 2-008). Center for the Improvement of Early Reading Achievement. <https://www.see-n-read.com/articles/2-008.pdf>
- Lochrin, M., Arciuli, J., & Sharma, M. (2015). Assessing the relationship between prosody and reading outcomes in children using the PEPS-C. *Scientific Studies of Reading*, 19(1), 72–85.
- Miller, J., & Schwanenflugel, P. J. (2006). Prosody of syntactically complex sentences in the oral reading of young children. *Journal of Educational Psychology*, 98(4), 839–853.
- Miller, J., & Schwanenflugel, P. J. (2008). A longitudinal study of the development of reading prosody as a dimension of oral reading fluency in early elementary school children. *Reading Research Quarterly*, 43(4), 336–354.
- Morrison, T. G., & Wilcox, B. (2020). Assessing expressive oral reading fluency. *Education in Science*, 10(3), 59. <https://doi.org/10.3390/educsci10030059>
- Myers, B. R., Lense, M. D., & Gordon, R. L. (2019). Pushing the envelope: Developments in neural entrainment to speech and the biological underpinnings of prosody perception. *Brain Sciences*, 9(3), 70. <https://doi.org/10.3390/brainsci9030070>
- Nese, J. F. T., Whitney, M., Alonzo, J., Sáez, L., & Nese, R. N. T. (2020). *Human-rated prosody study*. Retrieved from https://jnese.github.io/coreprosody/human_prosody_scoring.html
- Paige, D. D., Rasinski, T., Magpuri-Lavell, T., & Smith, G. S. (2014). Interpreting the relationships among prosody, automaticity, accuracy, and silent reading comprehension in secondary students. *Journal of Literacy Research*, 46(2), 123–156.
- Rasinski, T., Rikli, A., & Johnston, S. (2009). Reading fluency: More than automaticity? More than a concern for the primary grades? *Literacy Research & Instruction*, 48(4), 350–361.

- Rasinski, T. V. (2012). Why reading fluency should be hot! *The Reading Teacher*, 65(8), 516–522.
- Ríos-López, P., Molnar, M. T., Lizarazu, M., & Lallier, M. (2017). The role of slow speech amplitude envelope for speech processing and reading development. *Frontiers in Psychology*, 8, 1497. <https://doi.org/10.3389/fpsyg.2017.01497>
- Schreiber, P. A. (1991). Understanding prosody's role in reading acquisition. *Theory Into Practice*, 30, 158–164.
- Schremm, A., Horne, M., & Roll, M. (2015). Brain responses to syntax constrained by time-driven implicit prosodic phrases. *Journal of Neurolinguistics*, 35, 68–84.
- Schwanenflugel, P. J., & Benjamin, R. G. (2017). Lexical prosody as an aspect of oral reading fluency. *Reading and Writing*, 30, 143–162.
- Schwanenflugel, P. J., Hamilton, A. M., Kuhn, M. R., Wisenbaker, J. M., & Stahl, S. A. (2004). Becoming a fluent reader: Reading skill and prosodic features in the oral reading of young readers. *Journal of Educational Psychology*, 96(1), 119–129.
- Stahl, S. A., & Kuhn, M. R. (2002). Center for the Improvement of early Reading achievement: Making it sound like language: Developing fluency. *The Reading Teacher*, 55(6), 582–584.
- Tong, S. X., Tsui, R. K., Law, N. S. H., Fung, L. S. C., Chiu, M. M., & Cain, K. (2022). "And this one's juuust right!": Prosody predicts Reading comprehension in Hong Kong Chinese-English bilingual children. Unpublished manuscript. Downloaded from <https://www.psyarxiv.com>
- Veenendaal, N. J., Groen, M. A., & Verhoeven, L. (2016). Bidirectional relations between text reading prosody and reading comprehension in the upper primary school grades: A longitudinal perspective. *Scientific Studies of Reading*, 20(3), 189–202.
- Wade-Woolley, L., Wood, C., Chan, J., & Weidman, S. (2022). Prosodic competence as the missing component of reading processes across languages: Theory, evidence and future research. *Scientific Studies of Reading*, 26(2), 165–181.
- White, S., Sabatini, J., Park, B. J., Chen, J., Bernstein, J., & Li, M. (2021). *The 2018 NAEP Oral Reading fluency study (NCES 2021-025)*. U.S. Department of Education. Washington, DC: Institute of Education Sciences, National Center for Education Statistics. Retrieved from <https://nces.ed.gov/pubsearch/pubsinfo.asp?pubid=2021025>
- Wolters, A. P., Kim, Y. S. G., & Szura, J. W. (2022). Is reading prosody related to reading comprehension? A meta-analysis. *Scientific Studies of Reading*, 26(1), 1–20.
- Young, A. R., Bowers, P. G., & MacKinnon, G. E. (1996). Effects of prosodic modeling and repeated reading on poor readers' fluency and comprehension. *Applied PsychoLinguistics*, 17(1), 59–84.
- Young, C., Mohr, K. A., & Rasinski, T. (2015). Reading together: A successful reading fluency intervention. *Literacy Research & Instruction*, 54(1), 67–81.
- Young, C., & Rasinski, T. (2018). Readers theatre: Effects on word recognition automaticity and reading prosody. *Journal of Research in Reading*, 41(3), 475–485.

MORE TO EXPLORE

- Try this four-factor reading prosody scoring rubric in your classroom: http://www.timrasinski.com/presentations/multidimensional_fluency_rubric_4_factors.pdf
- For fluency resources (videos and website information) to help families promote prosody, visit: <https://www.readingrockets.org/reading-101-guide-parents/second-grade/fluency-activities-your-second-grader>
- To watch a teacher model and "think aloud" prosody during a classroom lesson, watch this brief YouTube video: https://www.youtube.com/watch?v=ulxBof_fgjo
- To learn more about how the National Assessment of Educational Progress (NAEP) is using reading prosody to score 4th-grade reading oral reading expression, read: https://nces.ed.gov/nationsreportcard/subject/studies/orf/2021025_orf_study.pdf